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Introduction

I'm a PhD student at the University of Edinburgh's Quantum Software Lab, funded by the Oxford Quantum Technology Hub DTP and VeriQloud, working on distributed quantum computing with Prof. Chris Heunen and Prof. Mahesh Marina. With a physics background, I've worked on quantum networking, photonic simulations, and quantum software across industry internships.

Work History

Quantum Software Lab

Edinburgh

PhD Researcher

Sep 2024 – Aug 2028

- Developing frameworks for model agnostic distributed quantum computing for hybrid workloads.
- Investigating circuit partitioning, communication protocols, and resource scheduling for quantum networks.

CQT

Singapore

Invited researcher

June 2026

- Invited researcher at the 2026 HACKMONTH.
- Investigated and published research related to quantum advantages in distributed learning and verification.

Quantinuum

Cambridge

Quantum compiler intern

Sep 2025 – Dec 2025

- Built a parallelized circuit optimizer in Rust for Quantinuum's TKET compiler.

National Quantum Computing Centre

STFC Harwell Campus

Quantum Software Resident

Mar 2024-May 2024

- Developed and deployed Trotterized Vlasov-Poisson simulations for plasma modelling on quantum hardware.
- Worked with NQCC application engineers to solve real-world use cases with quantum computers.

IBM Quantum

Zurich, Remote

Community Advocate Intern

May 2023 – Aug 2024

- Built and maintained IBM's UK Quantum Community, collaborating with partners and Quantum Hubs.
- Promoted IBMQ infrastructure adoption through tailored programs and community engagement.

Cisco

London

Placement Research Intern

Jul 2021 – Aug 2022

- Led internal quantum network simulation efforts using discrete event simulator tools.
- Collaborated with academic research groups from the Centre for Quantum Networks (CQN).

Google Summer of Code

Remote

Open Source Developer

Jun 2022 – Sep 2022

- Implemented a quantum error correction code library for Julia's QuantumClifford.jl Clifford gate simulator.

Education

PhD in Quantum Informatics

Sept 2024 – August 2028

University of Edinburgh

Conferences: SIGCOMM, ARC Quantum Summer School, FQCE25, Entanglement Information and Complexity in Quantum Systems, NQCC Scalability Conference, NPL Quantum Simulation and Digital Twins masterclass

BSc Physics

Sept 2019 – July 2024

University of Bath

Publications

Quantum ring all-reduce: communication and privacy advantages for distributed learning	arXiv 2026
On the Distortion of Partitioning Performance by Random Quantum Circuits	ICDCS DisQIC 2026
Distributed Quantum Circuit Optimisation: Evaluating Global and Local Encodings	ICDCS DisQIC 2026
Distributed Quantum Error Mitigation: Global and Local ZNE Encodings	INFOCOM QUNAP 2026
Distributed Quantum Computing Across Heterogeneous Hardware with Hybrid Dependency Hypergraphs	SIGCOMM Poster 2025
Introducing Quantum Computing to High-School Curricula: A Global Perspective	arXiv 2025
Controlling Quantum Communication via Quantum Memory Management	US Patent 2024
Realizing a Sustainable Quantum Internet for the Smallest/Future Researchers	QIH 2022

Awards

SandboxAQ Research Excellence Scholarship – \$10,000 award	Aug 2025
Unitary Foundation Microgrant – \$4,000 for HDH library development	Aug 2025
Femtum Leap Awards Finalist – Rising Star award nomination	2023
Prize Winner , Quantum Internet Hackathon 2022	2022

Selected Talks

Distributed Quantum Error Mitigation – <i>INFOCOM QUNAP</i>	Tokyo, Apr 2026
Distributed Quantum Circuit Optimisation – <i>QCTIP</i>	Oxford, Apr 2026
Hybrid Dependency Hypergraphs for Distributed Quantum Computing – <i>IWQC</i>	Helsinki, Sep 2025
Distributed Quantum Computing Frameworks – <i>WERQSHOP, NYU</i>	New York, Apr 2025
Quantum Software for the Next Generation – <i>Youth Quantum Summit</i>	Oxford, Nov 2024

Open Source Projects

HDH library (2025–present) – Python library for Hybrid Dependency Hypergraphs in distributed quantum computing.	Documentation
ZNE-DQC (2026) – Zero-noise extrapolation implementation for distributed quantum systems	GitHub
QEC library for QuantumClifford.jl (2022) – Comprehensive quantum error correction code library for Julia; Google Summer of Code project	GSoC summary

Service & Community

Peer Reviewer – IEEE Transactions on Quantum Engineering; EPJ Quantum Technology (Springer Nature; Asian Quantum Information Science Conference 2026)	2026 – present
Student Contributor , Physics World (IOP Publishing)	2026 – present
Ambassador , Classiq Technologies	2024 – 2028
Co-founder & Director , bqb Quantum Youth – international student-led quantum education initiative	2022 – 2026

Technical Skills

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- **Languages:** Python, Rust, C++, Julia
 - **Quantum frameworks:** Qiskit, pytket (TKET), NetSquid, SeQUeNCe
 - **Tools:** Git, GitHub Actions, L^AT_EX, HyperGraph partitioners (KaHyPar)